



PRODUCT NAME

High-performance Cold Cream Formula

Description

This high-performance cold cream offers excellent moisturizing and emollient properties while being non-comedogenic. It provides a cooling effect and maintains a rich texture, perfect for soothing and hydrating the skin.

Technical Specifications

pH Level:	7
Viscosity:	High
Shelf Life:	12 months
Batch Size:	10kg - 100kg
Processing Time:	2 hours
Temperature:	70°C to 40°C
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Homogenizer, heating vessels, mixing paddles
Certification:	GMP compliant

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	55.0%	Solvent and base for the emulsion
2	Mineral Oil	Paraffinum Liquidum	15.0%	Primary emollient with moisturizing benefits
3	Stearic Acid	Stearic Acid	8.0%	Thickener and co-emulsifier providing rich texture
4	Glycerin	Glycerin	7.0%	Humectant for enhancing moisture retention
5	Lanolin Alcohol	Lanolin Alcohol	6.0%	Emulsifier and emollient providing non-comedogenic properties
6	Coconut Oil	Cocos Nucifera Oil	4.0%	Emollient enhancing skin softness and cooling effect
7	Triethanolamine	Triethanolamine	1.5%	pH adjuster
8	Eucalyptus Oil	Eucalyptus Globulus Leaf Oil	1.0%	Cooling agent providing a refreshing sensation
9	Preservative	Phenoxyethanol	2.5%	Ensures product longevity and prevents microbial growth
Total			100%	

Manufacturing Process

Phase 1: Aqueous Phase

1. Heat Water (Aqua) to 70°C.
2. Add Glycerin and Triethanolamine to the water and mix until uniform.

Phase 2: Oil Phase

1. Heat Mineral Oil, Stearic Acid, and Lanolin Alcohol to 70°C.

2. Ensure complete melting and uniform mixing.

Phase 3: Emulsification

1. Slowly add the Oil Phase to Aqueous Phase under agitation.
2. Mix using a homogenizer for 10 minutes to achieve a stable emulsion.

Phase 4: Cooling Phase

1. Continue mixing while cooling to 40°C.
2. Add Coconut Oil, Eucalyptus Oil, and Preservative.
3. Mix until fully homogeneous.

Required Equipment

Homogenizer, heating vessels, mixing paddles

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: GMP compliant

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 10kg - 100kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply a small amount to cleansed skin, gently massaging until fully absorbed. Use daily for optimal moisturizing and soothing effects.